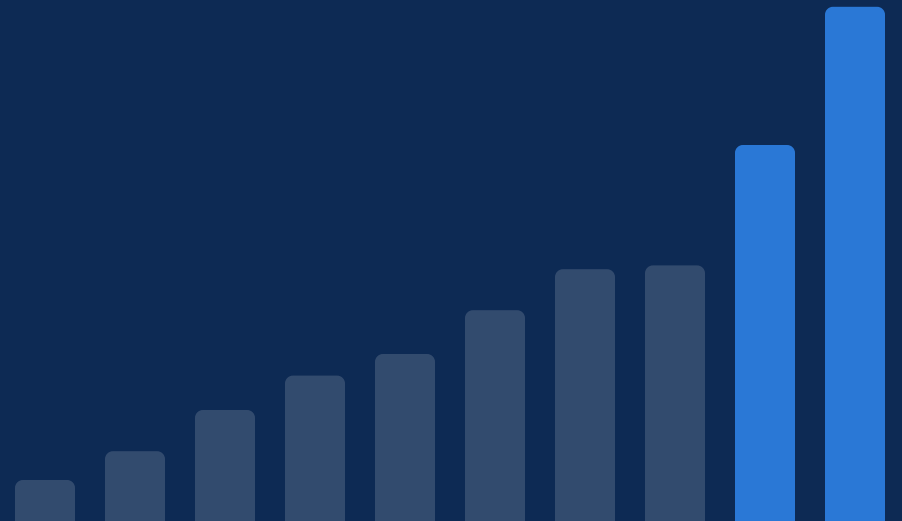

PREDICTIVE MODELING PROJECT

Which restaurants are likely to fail their next inspection?

A risk score for Las Vegas restaurants, built from inspection results and inspector comments

Adhrit Singh

October 2026



The risk score finds about twice as many future failures per visit

THE QUESTION

After an inspection, can we tell whether a restaurant's **next** inspection will be graded **C or below** (a "fail")?

HOW COMMON IS A FAIL?

About **1 in 4**: 23.7% of inspections are followed by a C or below.

WHY IT MATTERS

Loss-control teams can't visit every insured restaurant. A good ranking sends visits where problems are most likely.

41%

of future fails are in the **riskiest 20%** of scores: about **twice** the 20% that random visits would find.

47%

of the model's flags were right: the restaurant got a C or below next time, against **23%** on average.

73%

chance the model ranks a restaurant that will fail above one that won't. Random guessing scores 50%.

Good enough to prioritize visits, not to call any single restaurant. All three numbers come from 3,981 restaurants the model never saw while it was built.

35,000 inspections, cleaned, then split so the final test is fair

35,000

inspections

19,902

restaurants

2020–2026

April 2020 to September 2026

WHAT WAS IN IT, AND HOW IT WAS CLEANED

- Restaurant details, the latest inspection's **violations**, and the inspector's **written comments**, plus lookups for violation codes and inspector shorthand.
- Placeholders ("N/A", "TBD", "###"), misspellings and impossible values (e.g. a median staff age of 250) were fixed or marked missing.
- 34,708 map coordinates had lost their minus sign and were corrected.
- **No rows were dropped**; every cleaning step was logged and checked.

Restaurants were split before any analysis

Each restaurant is on one side only, just like scoring brand-new restaurants

Training · 80%

Test · 20%

15,921

restaurants

28,036 inspections

Used to **explore** the data,
build signals and **choose** the
model.

3,981

restaurants

6,964 inspections

Locked away and used **once**,
at the end, as the final exam.

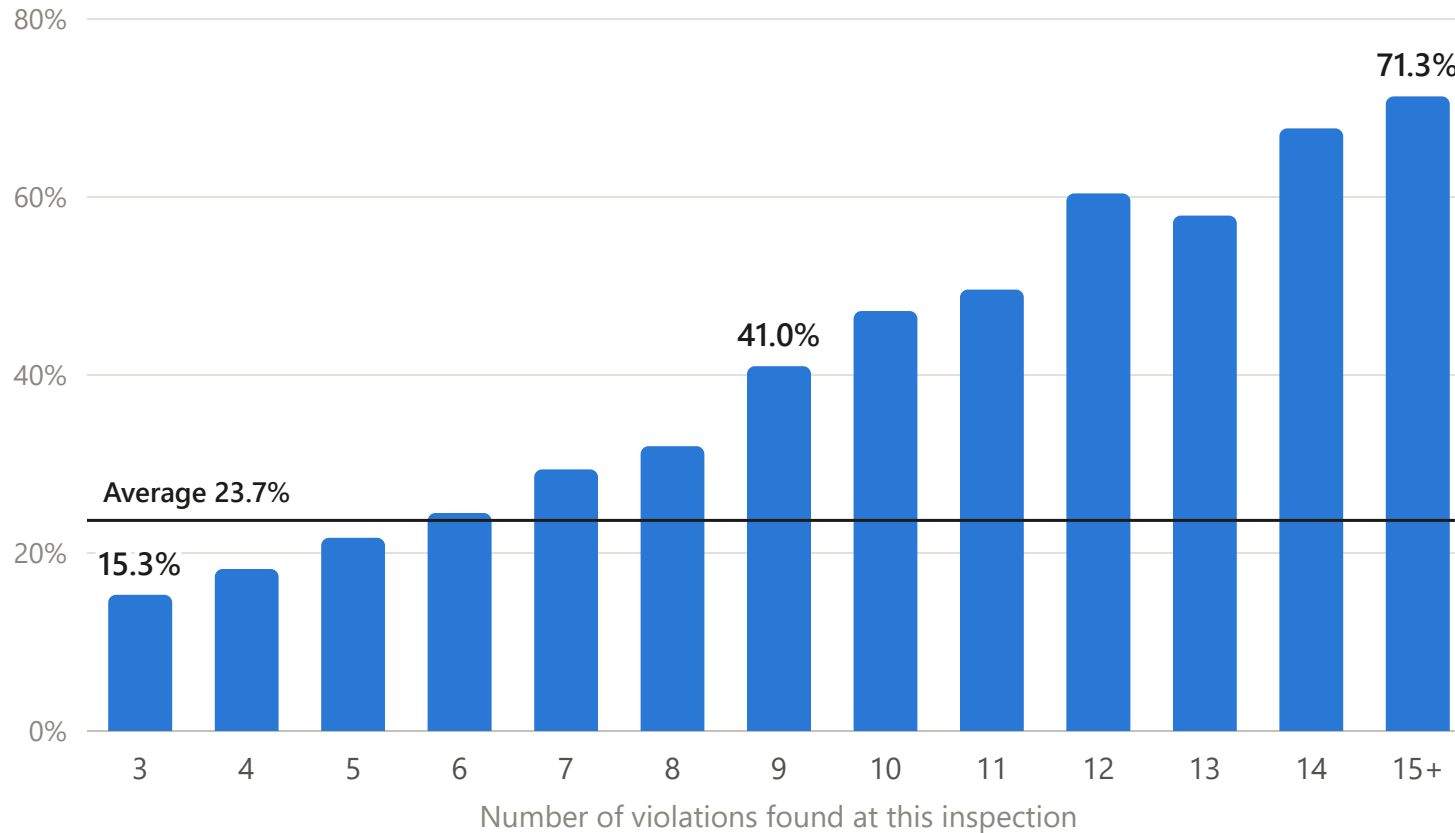
DEVELOPMENT TESTING

Within training, candidates were compared in **5 rounds**, each time testing on a different fifth of the training restaurants.

The strongest warning sign is what the latest inspection found

Share of inspections followed by a C or below, by number of violations

Training restaurants, 28,036 inspections. Every inspection had at least 3 violations.



STEADY CLIMB

From **15%** at 3 violations to **71%** at 15 or more, with no plateau.

CRITICAL VIOLATIONS

14% with none, rising to **58%** with five or more.

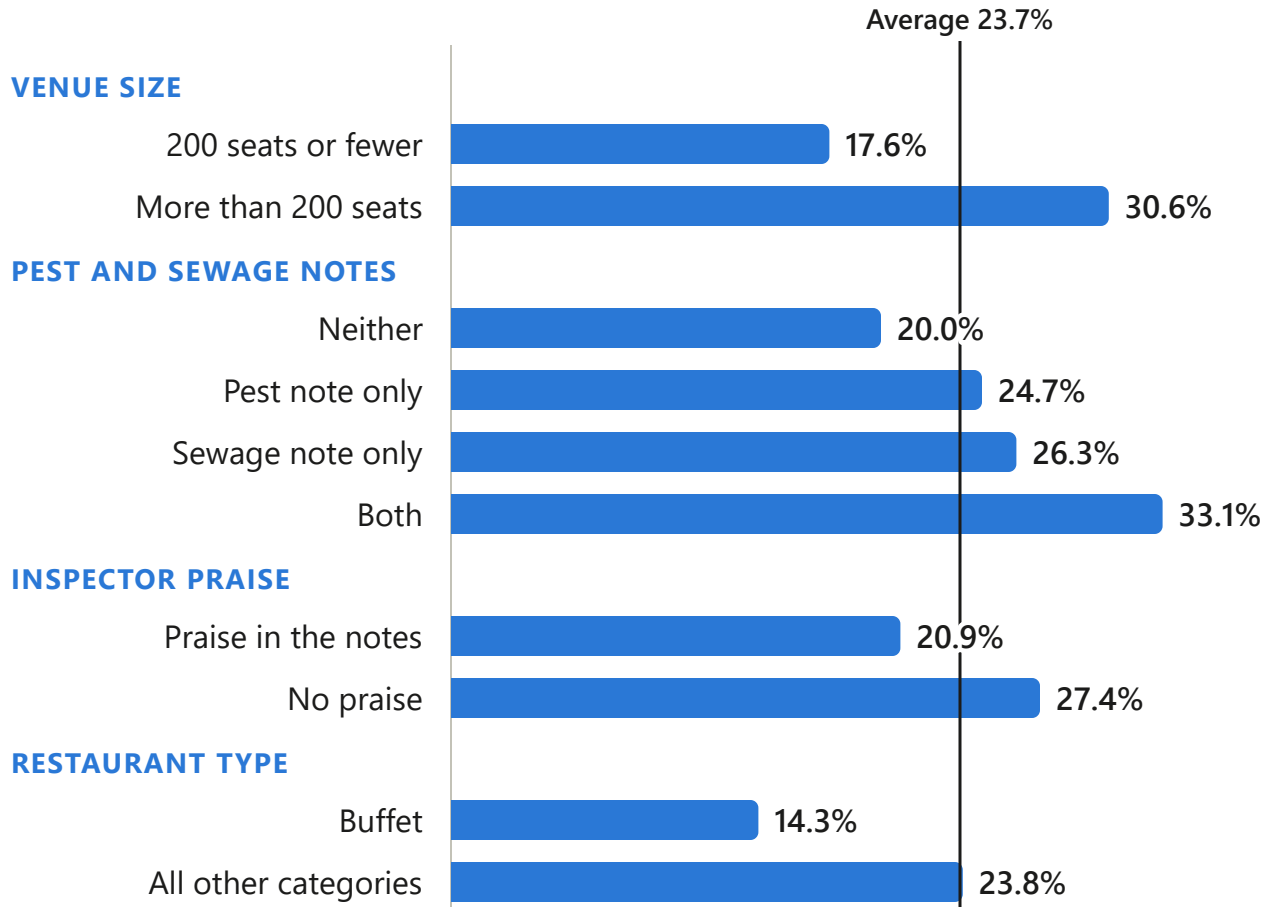
MOST SERIOUS VIOLATION

11% when the worst is minor, up to **30%** for an imminent health hazard.

Venue size and inspector notes add to the picture; many fields don't

Share followed by a C or below

Training restaurants. Associations, not causes.



DIDN'T HELP PREDICT RISK

- Customer complaints and foodborne-illness counts
- Review ratings, safety certification, allergen labeling
- Location (city, ZIP code)
- Staff numbers, age and tenure; restaurant age
- Opening hours, and the restaurant's earlier inspections
- Time: 21% to 26% in every quarter, no trend

Knowing what *doesn't* predict risk is useful too: these fields can be left out of a visit-planning tool.

From 49 candidate signals to the 11 that consistently help

1 Build 49 candidates

Eight groups: violations, restaurant details, inspector notes, earlier inspections, opening hours, business fields, current grade, and free text.

2 Read the inspector's comments

Expanded shorthand ("hws" → "handwash sink"), matched notes to food-safety regulation topics, and reviewed all **307** recurring note types by hand.

3 Keep only what helps reliably

A signal stayed only if it improved accuracy in at least **4 of 5** test rounds on the training restaurants.

Left out on purpose: the restaurant's current grade. It may have been recorded *after* the inspection, which could leak the answer, and it added nothing.

The final 11 signals

WHAT THE INSPECTION FOUND

5

Number of violations · number of critical violations · most serious violation · imminent health hazard · demerits of the first violation listed

THE RESTAURANT

2

More than 200 seats · buffet

THE INSPECTOR'S NOTES

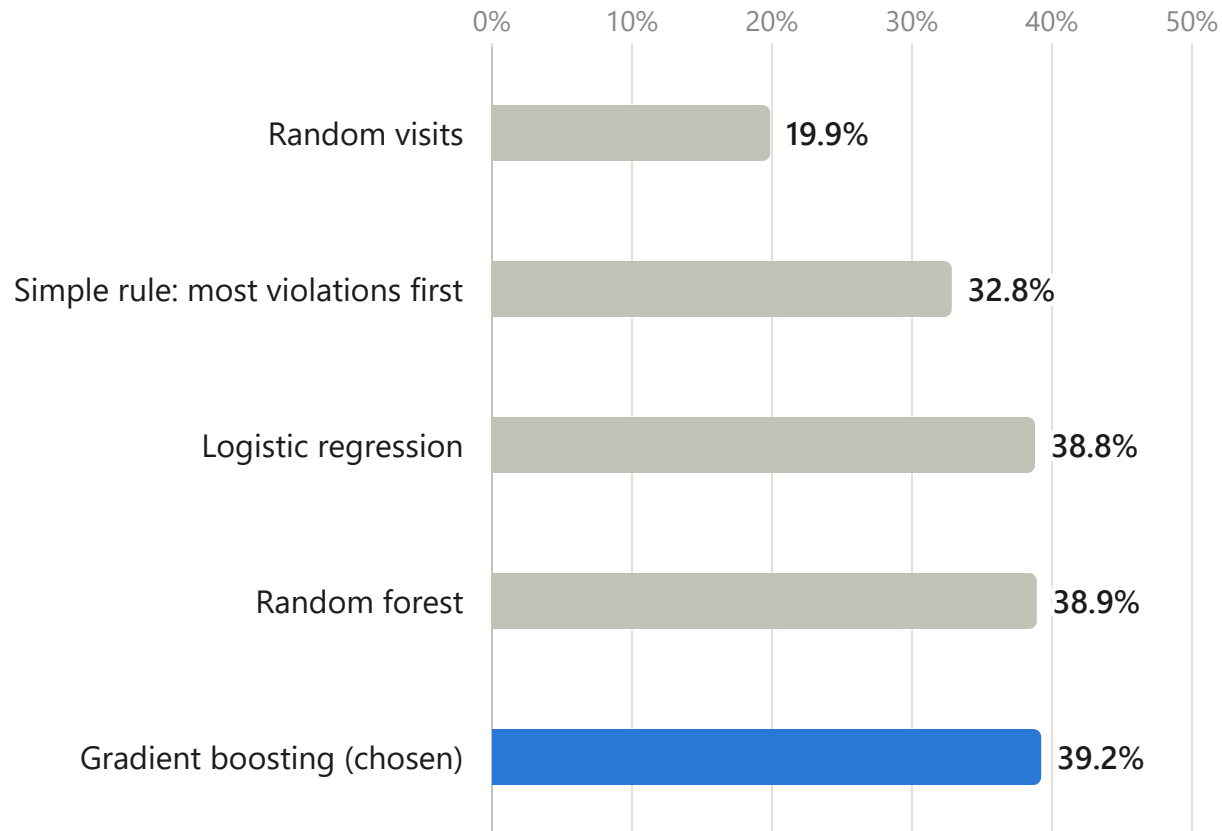
4

Pests · sewage · a critical item noted · praise for the kitchen

Three modeling approaches all beat simple rules; gradient boosting led

Share of future fails found in the riskiest 20% of the list

Development testing: 5 rounds, each holding out a different fifth of training restaurants



WHY GRADIENT BOOSTING

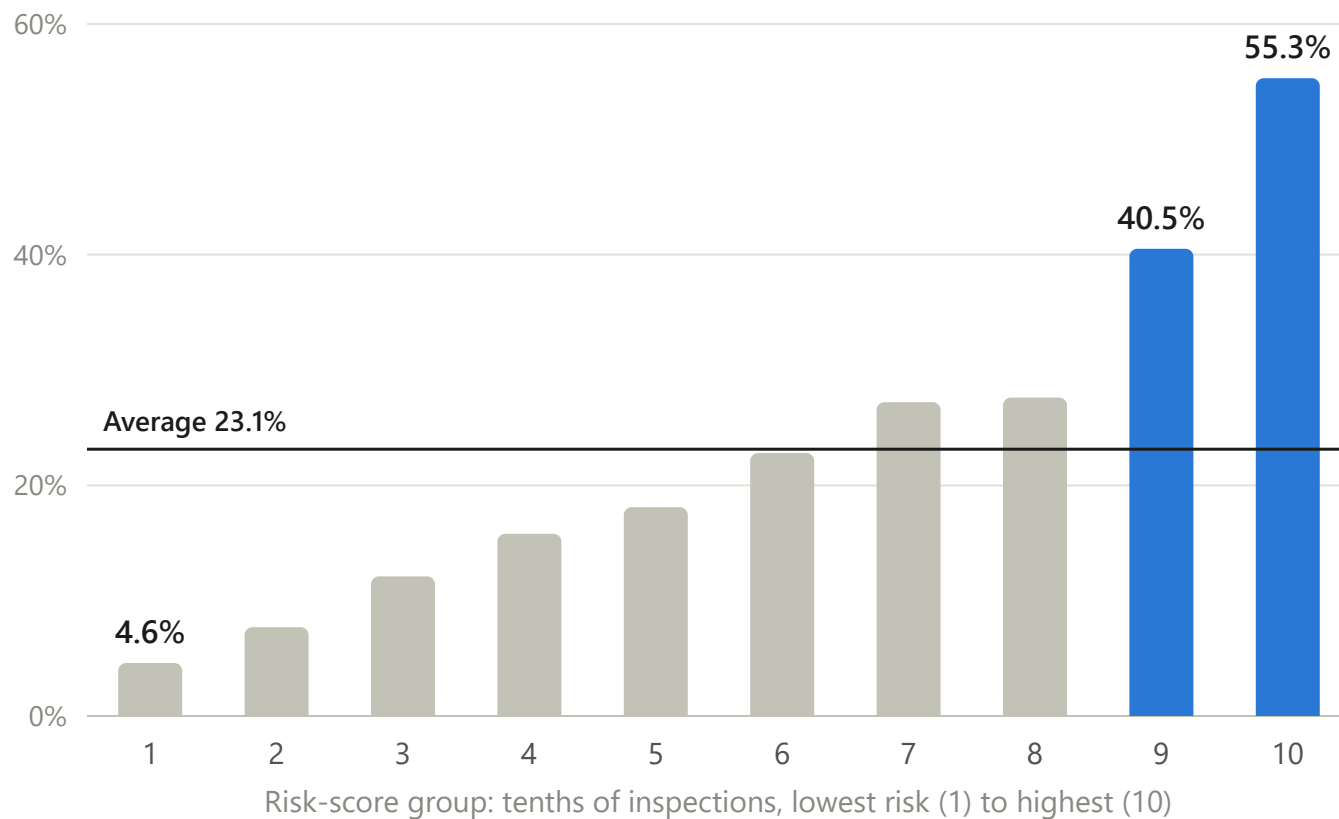
- **Most accurate:** ahead of logistic regression in all 5 rounds, and of the random forest in at least 4.
- **Captures combinations,** such as a large venue with many violations.
- **Explainable:** we can show what drove each score (slide 9).
- **Simple to maintain:** 11 inputs, standard tools.

The three models are close. If a transparent points-style score is preferred, logistic regression gives up very little.

On 3,981 restaurants it never saw, the score separates risk clearly

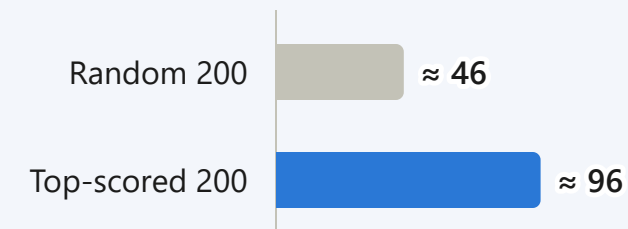
Share followed by a C or below, by risk-score group

■ Groups 1–8 ■ Top 20%: the visit list



PER 1,000 INSPECTIONS

About **231** are followed by a fail. Visiting 200 of them finds:



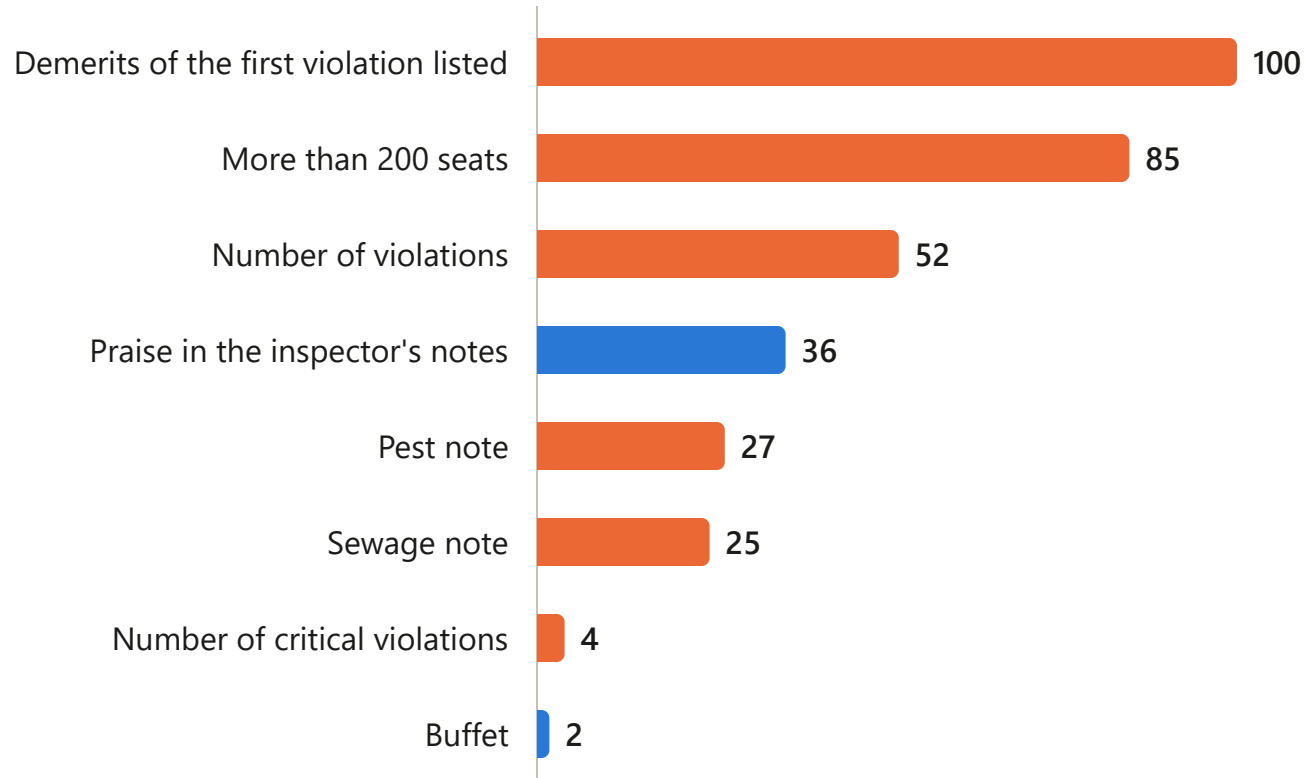
TWICE THE RETURN ON THE SAME EFFORT

The top 20% holds **41%** of future fails, in line with development testing (39%), so it holds up on restaurants it has never seen.

What pushes a restaurant's predicted risk up or down

Average influence on the score (largest = 100)

■ Raises risk ■ Lowers risk



Most serious violation, imminent health hazard and critical-item note scored near zero: their information overlaps with the signals above.

THE TOP THREE

How serious the **first violation listed** is, a venue with **more than 200 seats**, and the **number of violations**.

ASSOCIATIONS, NOT LEVERS

Praise marks a well-run kitchen; writing more praise wouldn't lower anyone's risk.

TO CONFIRM WITH DATA OWNERS

Why the *first* violation listed matters so much is undocumented.

RECOMMENDATIONS

Use the score to prioritize visits, and pilot it before relying on it

1 Rank, don't label

Give loss control a ranked list each period and work down it as capacity allows, e.g. the riskiest 20% first.

2 Read scores as group risk

A high score means "high priority", not "certain to fail".

3 Pilot on new inspections

Score inspections as they happen and compare with current practice and the simple violation-count rule.

4 Monitor and refresh

Track how many fails the list catches, the share flagged, and shifts in incoming data; refit on all data before launch.

5 Improve the data

Next-inspection dates, record timestamps and inspector IDs would allow a true forward-in-time test.

KNOW THE LIMITS

- **Not tested on the future.** The test used unseen restaurants from the same 2020–2026 period.
- **The data look partly synthetic:** inspections spread evenly around the clock and a perfectly clean 200-seat step. Real data may differ.
- **Timing can't be verified** for some fields; the current grade was excluded for that reason.
- **Single scores are less certain** than group averages.